

The Optimal Dynamic Mixture Weight is a Sigmoid: Continuous HJB and Discrete Soft-Bellman Derivations with Explicit Information Requirements

Abstract

An agent navigates by arbitrating between two innate stochastic behaviors, a geocentric (goal-directed) mode and an egocentric (obstacle-avoiding) mode, via a mixture weight $\gamma \in [0, 1]$. We derive, in continuous time via the Hamilton–Jacobi–Bellman equation and the stochastic maximum principle, and in discrete time via the soft (entropy-regularized) Bellman equation, that the optimal state-dependent mixture weight is in both cases the unique interior minimizer of a strictly convex Hamiltonian and is exactly a logistic sigmoid of a switching function. We then state precisely what evaluating that switching function requires — the exact state and the exact value function at that state — and prove that no memoryless reactive policy based on local observations can supply it. Local executable surrogates and memory-augmented switching functions are deliberately deferred to separate developments.

1 Part I: The Continuous-Time Problem

1.1 State, the two behaviors, and the noise

Assumption 1 (State and relaxed control). *The state is the position $\mathbf{r}_t = (x_t, y_t)^\top \in \Omega \subset \mathbb{R}^2$. The control is a relaxed (measure-valued) control: at each instant a behavioral mode $z_t \in \{G, E\}$ is drawn with $\mathbb{P}(z_t = G) = \gamma_t \in [0, 1]$. The two modes induce drift fields*

$$\mathbf{v}_G(x, y) = \begin{bmatrix} v_0 \\ \kappa \theta(x, y) \end{bmatrix}, \quad \mathbf{v}_E(x, y) = -c_E \nabla O(x, y) = \begin{bmatrix} -c_E \frac{\partial O}{\partial x}(x, y) \\ -c_E \frac{\partial O}{\partial y}(x, y) \end{bmatrix}, \quad c_E \equiv \frac{\alpha \epsilon^2}{2},$$

where v_0 is the forward speed, θ the goal-bearing deviation, κ the steering gain, O a smoothed obstacle field, α the avoidance strength and ϵ the microscopic step size. The effective drift is the convex combination

$$\mathbf{b}(\mathbf{r}, \gamma) = \gamma \mathbf{v}_G(\mathbf{r}) + (1 - \gamma) \mathbf{v}_E(\mathbf{r}),$$

i.e. componentwise

$$b_x(x, y, \gamma) = \gamma v_0 - (1 - \gamma) c_E \frac{\partial O}{\partial x}(x, y), \tag{1}$$

$$b_y(x, y, \gamma) = \gamma \kappa \theta(x, y) - (1 - \gamma) c_E \frac{\partial O}{\partial y}(x, y). \tag{2}$$

Assumption 2 (Environment-supplied noise). *The noise is exogenous and control-independent:*

$$d\mathbf{r}_t = \mathbf{b}(\mathbf{r}_t, \gamma_t) dt + \mathbf{B} d\mathbf{W}_t, \quad \mathbf{B} = \begin{bmatrix} \sqrt{2D_{\parallel}} & 0 \\ 0 & \sqrt{2D_{\perp}} \end{bmatrix}, \quad \mathbf{D} = \frac{1}{2} \mathbf{B} \mathbf{B}^\top = \begin{bmatrix} D_{\parallel} & 0 \\ 0 & D_{\perp} \end{bmatrix} \succ 0,$$

with $\mathbf{W}$ a standard 2D Wiener process and $\mathbf{D}$ constant in the working frame. Componentwise,

$$dx_t = \left[\gamma_t v_0 - (1 - \gamma_t) c_E \frac{\partial O}{\partial x}(x_t, y_t) \right] dt + \sqrt{2D_{\parallel}} dW_{x,t}, \quad (3)$$

$$dy_t = \left[\gamma_t \kappa \theta(x_t, y_t) - (1 - \gamma_t) c_E \frac{\partial O}{\partial y}(x_t, y_t) \right] dt + \sqrt{2D_{\perp}} dW_{y,t}. \quad (4)$$

Note $\mathbb{E}[\mathbf{B} d\mathbf{W}_t] = \mathbf{0}$: anisotropy resides in the covariance $\mathbb{E}[(\mathbf{B} d\mathbf{W})(\mathbf{B} d\mathbf{W})^{\top}] = 2\mathbf{D} dt$, never in the mean.

Assumption 3 (Diffusive limit and the Girsanov hypothesis). *The SDE above is the Kramers–Moyal limit of the discrete mixture random walk with step ϵ and time step Δt , $\epsilon^2 \propto \Delta t$, with smooth tilts, so that per-step probability deviations from the passive kernel are $O(\sqrt{\Delta t})$. Hence the mean displacement is $O(\Delta t)$ (a finite drift) while the covariance is $2\mathbf{D} \Delta t + o(\Delta t)$ identically for passive and controlled kernels; the two path measures differ only in drift and are equivalent (Girsanov). A control acting on the covariance at order Δt would render the measures mutually singular (infinite KL rate) and is excluded.*

1.2 Walls, goal, and boundary conditions

Assumption 4 (Two obstacle channels and boundary data). *Obstacles act through (E) environmental rejection of blocked moves, whose continuum limit is a no-flux boundary, and (P) policy-level repulsion through the smoothed field O in $\mathbf{v}_E$. Let $\rho(\mathbf{r}, t)$ be the probability density and $\mathbf{n} = (n_x, n_y)$ the outward normal on wall boundaries $\partial\Omega_w$. For every γ :*

$$(\text{Fokker–Planck, no flux}) \quad n_x \left[b_x \rho - D_{\parallel} \frac{\partial \rho}{\partial x} \right] + n_y \left[b_y \rho - D_{\perp} \frac{\partial \rho}{\partial y} \right] = 0 \quad \text{on } \partial\Omega_w, \quad (5)$$

$$(\text{HJB, Neumann}) \quad n_x \frac{\partial V}{\partial x} + n_y \frac{\partial V}{\partial y} = 0 \quad \text{on } \partial\Omega_w, \quad (6)$$

$$(\text{goal, absorbing}) \quad V = 0 \quad \text{at the goal.} \quad (7)$$

The Neumann condition removes the boundary local-time term in Itô–Tanaka. Without the no-flux condition the model at $\gamma = 1$ is identically free of O and predicts unit success rate; the condition restores the trapped mass.

1.3 Running cost

Assumption 5 (Running cost, derived rather than postulated). *The cost is time plus the price of randomization. In discrete time the maximum-entropy objective carries a per-step entropy bonus $\eta \Delta t H_b(\gamma)$ with $H_b(\gamma) = -[\gamma \ln \gamma + (1 - \gamma) \ln(1 - \gamma)]$; dividing by Δt gives the continuous rate. Equivalently, charging the KL divergence of the joint (mode, move) law against a reference (γ_0, p_0) yields the Bernoulli KL, which differs from the entropy form only by a constant and a linear shift $\ln \frac{\gamma_0}{1 - \gamma_0}$ in the switching function. Thus*

$$L(\gamma) = 1 - \eta H_b(\gamma) = 1 + \eta \left[\gamma \ln \gamma + (1 - \gamma) \ln(1 - \gamma) \right], \quad \eta > 0.$$

The parameter η is free (the price of randomness). The channel sharpnesses α, β are dimensionless competences; no identity of the form $\eta = 1/\alpha + 1/\beta$ is imposed.

Remark 1.1 (The Itô correction). *Applying Itô’s lemma to $V(\mathbf{r}_t)$ contributes $D_{\parallel} \frac{\partial^2 V}{\partial x^2} dt + D_{\perp} \frac{\partial^2 V}{\partial y^2} dt$ from quadratic variation, since $(dx)^2 = 2D_{\parallel} dt$, $(dy)^2 = 2D_{\perp} dt$, $dx dy = 0$. This term remains in the HJB equation, in the Fokker–Planck equation and in the adjoint equation; it drops only from $\arg \min_{\gamma}$, because it contains no γ . Since $\mathbf{D}$ is constant, the spurious drift $\nabla \cdot \mathbf{D}$ vanishes.*

1.4 Value function, Hamiltonian, and the HJB equation

Let $V(\mathbf{r})$ be the optimal cost-to-go (expected time plus randomness price until absorption at the goal) and define the price vector

$$p_x = \frac{\partial V}{\partial x}(x, y), \quad p_y = \frac{\partial V}{\partial y}(x, y).$$

The Hamilton–Jacobi–Bellman equation is

$$0 = \min_{0 \leq \gamma \leq 1} \left\{ 1 + \eta [\gamma \ln \gamma + (1 - \gamma) \ln(1 - \gamma)] + p_x [\gamma v_0 - (1 - \gamma) c_E \frac{\partial O}{\partial x}] + p_y [\gamma \kappa \theta - (1 - \gamma) c_E \frac{\partial O}{\partial y}] + D_{\parallel} \frac{\partial^2 V}{\partial x^2} + D_{\perp} \frac{\partial^2 V}{\partial y^2} \right\}, \quad (8)$$

with the boundary data of Assumption 4. The control-dependent part is the *Hamiltonian*,

$$H(x, y, \gamma, p_x, p_y) = 1 + \eta [\gamma \ln \gamma + (1 - \gamma) \ln(1 - \gamma)] + p_x [\gamma v_0 - (1 - \gamma) c_E \frac{\partial O}{\partial x}] + p_y [\gamma \kappa \theta - (1 - \gamma) c_E \frac{\partial O}{\partial y}], \quad (9)$$

readable as an instantaneous price tag: price of time, plus price of randomizing, plus price of motion ($\mathbf{p} \cdot \mathbf{b}$, the rate at which moving changes the remaining cost).

Remark 1.2 (Stochastic maximum principle and verification). *Because $\mathbf{B}$ is control-independent, the stochastic maximum principle applies with first adjoint $\mathbf{p}_t = \nabla V(\mathbf{r}_t)$ and second adjoint $\mathbf{q}_t = \mathbf{B}^\top \nabla^2 V(\mathbf{r}_t)$ satisfying the backward SDE $d\mathbf{p}_t = -\nabla_{\mathbf{r}} [L + \mathbf{b} \cdot \mathbf{p} + \text{Tr}(\mathbf{D} \nabla^2 V)](\mathbf{r}_t) dt + \mathbf{q}_t d\mathbf{W}_t$, with transversality given by the boundary data. The necessary condition is $\gamma_t = \arg \min_{\gamma} H(\mathbf{r}_t, \gamma, \mathbf{p}_t)$ almost everywhere; since $\partial \mathbf{B} / \partial \gamma \equiv 0$, the second adjoint never enters this minimization. Any $C^2(\bar{\Omega})$ solution of (8) with the stated boundary conditions certifies the minimizing policy as globally optimal (verification theorem).*

1.5 Pointwise minimization: the sigmoid law

Theorem 1.3 (Unique interior soft bang-bang minimizer, continuous case). *For each (x, y, p_x, p_y) , the Hamiltonian (9) admits a unique minimizer in $(0, 1)$, given by*

$$\gamma^*(x, y) = \sigma\left(-\frac{\Delta H(x, y)}{\eta}\right), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with switching function

$$\Delta H(x, y) = p_x \left(v_0 + c_E \frac{\partial O}{\partial x} \right) + p_y \left(\kappa \theta + c_E \frac{\partial O}{\partial y} \right).$$

Proof. Differentiate (9) term by term. Using $\frac{d}{d\gamma} [\gamma \ln \gamma] = \ln \gamma + 1$ and $\frac{d}{d\gamma} [(1 - \gamma) \ln(1 - \gamma)] = -\ln(1 - \gamma) - 1$:

$$\begin{aligned} \frac{\partial H}{\partial \gamma} &= \eta (\ln \gamma + 1) - \eta (\ln(1 - \gamma) + 1) + p_x v_0 + p_x c_E \frac{\partial O}{\partial x} + p_y \kappa \theta + p_y c_E \frac{\partial O}{\partial y} \\ &= \eta \ln \frac{\gamma}{1 - \gamma} + \Delta H(x, y). \end{aligned}$$

Setting this to zero: $\ln \frac{\gamma}{1 - \gamma} = -\frac{\Delta H}{\eta}$, hence $\frac{\gamma}{1 - \gamma} = e^{-\Delta H/\eta}$. Writing $E = e^{-\Delta H/\eta}$, $\gamma = E/(1 + E)$, so $\gamma(1 + E) = E$ and $\gamma^* = \frac{E}{1 + E} = \frac{1}{1 + e^{\Delta H/\eta}} = \sigma(-\Delta H/\eta)$. The stationary point is the global minimizer because $\frac{\partial^2 H}{\partial \gamma^2} = \eta \left(\frac{1}{\gamma} + \frac{1}{1 - \gamma} \right) = \frac{\eta}{\gamma(1 - \gamma)} > 0$, and it is strictly interior because $\frac{\partial H}{\partial \gamma} \rightarrow -\infty$ as $\gamma \rightarrow 0^+$ and $\rightarrow +\infty$ as $\gamma \rightarrow 1^-$. $\square$

Substituting the minimizer back into (8) yields the closed-form (log-sum-exp, soft-min) HJB equation

$$0 = 1 + D_{\parallel} \frac{\partial^2 V}{\partial x^2} + D_{\perp} \frac{\partial^2 V}{\partial y^2} - \eta \ln \left(e^{-\frac{1}{\eta} \mathbf{p} \cdot \mathbf{v}_G} + e^{-\frac{1}{\eta} \mathbf{p} \cdot \mathbf{v}_E} \right),$$

with $\mathbf{p} \cdot \mathbf{v}_G = p_x v_0 + p_y \kappa \theta$ and $\mathbf{p} \cdot \mathbf{v}_E = -c_E \left(p_x \frac{\partial O}{\partial x} + p_y \frac{\partial O}{\partial y} \right)$. Limits: $\eta \rightarrow 0$ recovers hard bang-bang control; $\eta \rightarrow \infty$ gives $\gamma^* \rightarrow \frac{1}{2}$ (or the reference occupancy γ_0 under the KL-to-reference reading).

1.6 What the continuous law requires

Proposition 1.4 (Information requirement, continuous case). *Evaluating $\gamma^*(x, y)$ requires the six ingredients v_0, κ, c_E, η (self-knowledge), $\theta(x, y)$ (goal bearing), $\frac{\partial O}{\partial x}, \frac{\partial O}{\partial y}$ (local wall slopes), and p_x, p_y (the local prices). The prices are not local objects: V solves the boundary-value problem (8) over all of Ω , so ∇V at a point compresses the geometry of the entire domain. Consequently there exist pairs of states $\mathbf{r}_A, \mathbf{r}_B$ with identical local observations (same goal bearing, same local wall slopes) but $\nabla V(\mathbf{r}_A) \neq \nabla V(\mathbf{r}_B)$ and hence $\gamma^*(\mathbf{r}_A) \neq \gamma^*(\mathbf{r}_B)$. Any memoryless reactive policy, being a function of the current observation alone, outputs the same value at $\mathbf{r}_A$ and $\mathbf{r}_B$ and is therefore suboptimal at one of them. The sigmoid law is thus the full-information performance ceiling, not an executable controller for locally observing agents.*

2 Part II: The Discrete-Time Problem

2.1 Grid, actions, and the two stochastic behaviors

States are cells $s = (x, y)$, $x, y \in \{0, \dots, N-1\}$. The four actions and their step vectors are

$$a_N = (0, -1), \quad a_S = (0, +1), \quad a_E = (+1, 0), \quad a_W = (-1, 0).$$

The move rule encodes the wall physics: with $O(s') = 1$ if s' is a wall or outside the grid and 0 otherwise,

$$s'_a = \begin{cases} s + a & \text{if } O(s + a) = 0, \\ s & \text{if } O(s + a) = 1. \end{cases}$$

The stay-put rule is the discrete no-flux boundary condition; it holds for every γ automatically. The goal cell is absorbing with $V(\text{goal}) = 0$.

With unit compass direction to the goal (u_x, u_y) at s , the geocentric policy is the softmax over action-compass alignments,

$$\pi_G(a|s) = \frac{\exp(\beta(a_x u_x + a_y u_y))}{Z_G(s)}, \quad Z_G(s) = e^{-\beta u_y} + e^{\beta u_y} + e^{\beta u_x} + e^{-\beta u_x},$$

and the egocentric policy is the softmax over one-step obstacle penalties,

$$\pi_E(a|s) = \frac{\exp(-\alpha O(s+a))}{Z_E(s)}, \quad Z_E(s) = \sum_{a \in \{a_N, a_S, a_E, a_W\}} e^{-\alpha O(s+a)}.$$

The mixture policy is $\pi_\gamma(a|s) = \gamma \pi_G(a|s) + (1 - \gamma) \pi_E(a|s)$.

2.2 Meta-values and the soft Bellman equation

The per-step cost is $1 + \eta_d[\gamma \ln \gamma + (1 - \gamma) \ln(1 - \gamma)]$ with per-step temperature $\eta_d > 0$. Define the one-step-committed meta-values (commit to one behavior for this step, behave optimally thereafter):

$$Q_G(s) = 1 + \sum_a \pi_G(a|s) V(s'_a), \quad (10)$$

$$Q_E(s) = 1 + \sum_a \pi_E(a|s) V(s'_a). \quad (11)$$

Because Q_G, Q_E do not depend on γ , the Hamiltonian is affine in γ plus the entropy price, exactly as in the continuous case. The Bellman equation is

$$V(s) = \min_{0 \leq \gamma \leq 1} \left\{ \eta_d[\gamma \ln \gamma + (1 - \gamma) \ln(1 - \gamma)] + \gamma Q_G(s) + (1 - \gamma) Q_E(s) \right\}, \quad V(\text{goal}) = 0. \quad (12)$$

Performing the minimization (Theorem below) turns (12) into the closed-form *soft (entropy-regularized) Bellman equation* in cost convention,

$$V(s) = -\eta_d \ln \left(e^{-Q_G(s)/\eta_d} + e^{-Q_E(s)/\eta_d} \right), \quad (13)$$

i.e. a soft-min (log-sum-exp) over the two meta-values. Under the reward convention $V_R = -V$, $Q_R = -Q$, equation (13) reads $V_R(s) = \eta_d \ln(e^{Q_R(s,G)/\eta_d} + e^{Q_R(s,E)/\eta_d})$, the familiar soft Bellman (log-sum-exp) form. The Bellman operator is a contraction, so value iteration converges to the unique fixed point and the resulting law is exact, with no smoothness assumptions.

2.3 Pointwise minimization: the sigmoid law

Theorem 2.1 (Unique interior soft bang-bang minimizer, discrete case). *For each cell s , the objective in (12) admits a unique minimizer in $(0, 1)$,*

$$\gamma^*(s) = \sigma\left(\frac{Q_E(s) - Q_G(s)}{\eta_d}\right), \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Proof. The derivative with respect to γ is $\eta_d(\ln \gamma + 1) - \eta_d(\ln(1 - \gamma) + 1) + Q_G(s) - Q_E(s) = \eta_d \ln \frac{\gamma}{1 - \gamma} + (Q_G(s) - Q_E(s))$. Setting it to zero gives $\frac{\gamma}{1 - \gamma} = e^{-(Q_G - Q_E)/\eta_d}$, whence $\gamma^* = \frac{1}{1 + e^{(Q_G - Q_E)/\eta_d}} = \sigma((Q_E - Q_G)/\eta_d)$. Strict convexity, $\frac{\partial^2}{\partial \gamma^2} = \frac{\eta_d}{\gamma(1 - \gamma)} > 0$, and infinite slope at both endpoints give uniqueness and interiority exactly as in the continuous case. $\square$

In words: the optimal coin favors whichever behavior has the smaller total price (one step plus optimal continuation), smoothly, with temperature η_d . The limit $\eta_d \rightarrow 0$ recovers hard bang-bang, $\gamma^* \rightarrow \mathbf{1}[Q_E > Q_G]$.

2.4 What the discrete law requires

Proposition 2.2 (Information requirement, discrete case). *Evaluating $\gamma^*(s)$ requires $Q_G(s) - Q_E(s) = \sum_a (\pi_G(a|s) - \pi_E(a|s)) V(s'_a)$, and since $\sum_a (\pi_G(a|s) - \pi_E(a|s)) = 0$, this equals $\sum_a (\pi_G(a|s) - \pi_E(a|s)) (V(s'_a) - V(s))$: the value levels cancel, but the value differences at the neighboring cells do not, and these differences encode the geometry of the entire maze beyond the neighborhood. Hence there exist cells s_A, s_B with identical observations (same compass bearing, same four neighbor occupancies) but $Q_G(s_A) - Q_E(s_A) \neq Q_G(s_B) - Q_E(s_B)$. Any memoryless reactive policy outputs the same γ at both and is suboptimal at one. The discrete sigmoid law is likewise a full-information ceiling.*

3 Part III: The Dictionary Between the Two Frameworks

Expand the discrete switching signal. With $V(s'_a) - V(s) \approx \Delta t (a/\Delta t) \cdot \nabla V$ under the diffusive scaling of Assumption 3, and $\sum_a \pi_G(a|s) a = \mathbf{v}_G \Delta t$, $\sum_a \pi_E(a|s) a = \mathbf{v}_E \Delta t$:

$$Q_G(s) - Q_E(s) = \sum_a (\pi_G(a|s) - \pi_E(a|s)) (V(s'_a) - V(s)) \approx \Delta t (\mathbf{v}_G - \mathbf{v}_E) \cdot \nabla V = \Delta t \Delta H.$$

Matching $\sigma(-(Q_G - Q_E)/\eta_d)$ with $\sigma(-\Delta H/\eta)$ forces the temperature dictionary

$$\eta_d = \eta \Delta t$$

and identifies the discrete switching signal as the continuous one per unit step. Table 1 summarizes the correspondence.

Table 1: Dictionary between the continuous and discrete frameworks.

Object	Continuous	Discrete
State	$\mathbf{r} \in \Omega$	cell s
Control	$\gamma_t \in [0, 1]$ (relaxed)	γ (meta-coin)
Drift / kernel	$\mathbf{b} = \gamma \mathbf{v}_G + (1 - \gamma) \mathbf{v}_E$	$\pi_\gamma = \gamma \pi_G + (1 - \gamma) \pi_E$
Noise	explicit $\mathbf{B} d\mathbf{W}$, $\mathbf{B}\mathbf{B}^\top = 2\mathbf{D}$	implicit in the transition kernel
Diffusion term	explicit $\text{Tr}(\mathbf{D}\nabla^2 V)$	implicit in $\sum_a \pi(a s) V(s'_a)$
Walls	imposed: no-flux FPE, Neumann HJB	automatic: stay-put on blocked moves
Goal	absorbing, $V = 0$	absorbing, $V = 0$
Temperature	η (per unit time)	$\eta_d = \eta \Delta t$ (per step)
Switching signal	$\Delta H = \mathbf{p} \cdot (\mathbf{v}_G - \mathbf{v}_E)$	$Q_G - Q_E = \Delta t \Delta H + o(\Delta t)$
Optimal law	$\gamma^* = \sigma(-\Delta H/\eta)$	$\gamma^* = \sigma((Q_E - Q_G)/\eta_d)$
Master equation	HJB (8) / soft-min form	soft Bellman (12), (13)

4 Part IV: Similarities, Differences, and Conclusion

Similarities.

1. *Identical variational structure.* In both frameworks the Hamiltonian is affine in γ plus the Bernoulli entropy price; strict convexity with infinite endpoint slope yields a unique interior minimizer, and that minimizer is exactly the logistic sigmoid of a switching function. Soft bang-bang is thus structural, not model-specific.
2. *Identical meaning of the switching signal.* ΔH and $Q_E - Q_G$ are the same object: the price difference between committing to the geocentric versus the egocentric behavior for one instant (one step), with optimal continuation.
3. *Identical entropy price.* Both temperatures descend from the same per-step entropy bonus; the dictionary $\eta_d = \eta \Delta t$ makes the two laws agree order by order.
4. *Identical information obstruction.* In both frameworks the switching signal depends on value differences beyond the local observation, so aliasing pairs exist and no memoryless reactive policy attains the law. Both sigmoids are full-information ceilings.
5. *Identical limits.* $\eta \rightarrow 0$ (resp. $\eta_d \rightarrow 0$) recovers hard bang-bang; $\eta \rightarrow \infty$ (resp. $\eta_d \rightarrow \infty$) recovers the indifferent coin.

Differences.

1. *Exactness.* The discrete law is exact: the Bellman operator is a contraction and value iteration converges to the true fixed point. The continuous law is exact only subject to the diffusive-limit and regularity assumptions (Assumptions 2–4) and a verification argument.
2. *Where the noise lives.* Discretely, diffusion is implicit in the neighborhood expectation $\sum_a \pi(a|s)V(s'_a)$; continuously it must be supplied explicitly as $\mathbf{B}$, together with the Girsanov hypothesis (Assumption 3) that justifies the drift-only control and the quadratic information cost.
3. *Where the walls live.* Discretely, the stay-put rule is an automatic no-flux boundary for every γ ; continuously the no-flux and Neumann conditions must be imposed by hand, and omitting them produces the $\gamma = 1$ artifact.
4. *Value levels.* Discretely, the invariance of the switching signal to constant shifts of V is manifest (the weights sum to zero); continuously it appears as the statement that only ∇V enters.
5. *Second-order term.* The Itô correction $\text{Tr}(\mathbf{D}\nabla^2 V)$ is an explicit object continuously and an implicit shadow of the neighborhood expectation discretely.

Conclusion. Under Assumptions 1–5, the optimal dynamic mixture weight between two innate stochastic navigation behaviors is, in continuous time and in discrete time alike, the unique interior minimizer of a strictly convex Hamiltonian and is exactly $\gamma^* = \sigma(-\Delta H/\eta) = \sigma((Q_E - Q_G)/\eta_d)$: a smooth (soft bang-bang) switch whose argument is the price difference between the two behaviors. Evaluating that argument requires the exact state and the exact value function (equivalently its local differences) at that state; since these are global objects and locally aliased states demand different switches, the law is a performance ceiling rather than an executable controller for memoryless, locally observing agents. Executable local surrogates and memory-augmented switching functions are developed separately.